

## Laboratorium Informatika

### Junior Project (Teknik Basis Data, Pemrograman Berbasis Objek)

### Worksheet Responsi

#### RESPONSI

#### TUGAS WAJIB

Link Github: <https://github.com/rhizal24/514719-Muhammad-Rhizal-Rhomadon-ResponsiJuniorProject>

Screenshot file Tampilan:

Form1



#### DTETI Tracker

Developer Team Performance Tracker

##### DATA DEVELOPER

Nama Developer :   
Pilih Proyek :   
Status Kontrak :

##### DATA KINERJA

Fitur Selesai :  (Jumlah fitur yang dikerjakan)  
Jumlah Bug :  (Jumlah bug yang ditemukan)

INSERT

UPDATE

DELETE

##### DATA PERFORMA TIM

|   | Nama             | Proyek           | Status    | Fitur | Bug | Skor   | TOTAL GAJI |
|---|------------------|------------------|-----------|-------|-----|--------|------------|
| ▶ | Rudi WebDev      | Web Company ...  | Freelance | 5     | 1   | 86,67  | 2.500.000  |
|   | Siti Backend     | Sistem Parkir QR | Full Time | 12    | 3   | 105,00 | 7.100.000  |
|   | Budi Android     | Sistem Parkir QR | Freelance | 8     | 2   | 83,33  | 4.000.000  |
|   | Eko IoT Engineer | IoT Agriculture  | Full Time | 20    | 5   | 175,00 | 8.500.000  |

Screenshot Konsep OOP (Enkapsulasi dan *Inheritance*):

## Encapsulation

```
public class Developer : BaseModel
{
    // Private fields
    private string _namaDeveloper = string.Empty;
    private int _fiturSelesai;
    private double _skorTotal;

    // Public properties dengan validasi
    public string NamaDeveloper
    {
        get { return _namaDeveloper; }
        set { _namaDeveloper = value ?? string.Empty; }
    }

    public int FiturSelesai
    {
        get { return _fiturSelesai; }
        set { _fiturSelesai = value >= 0 ? value : 0; } // Validasi nilai negatif
    }

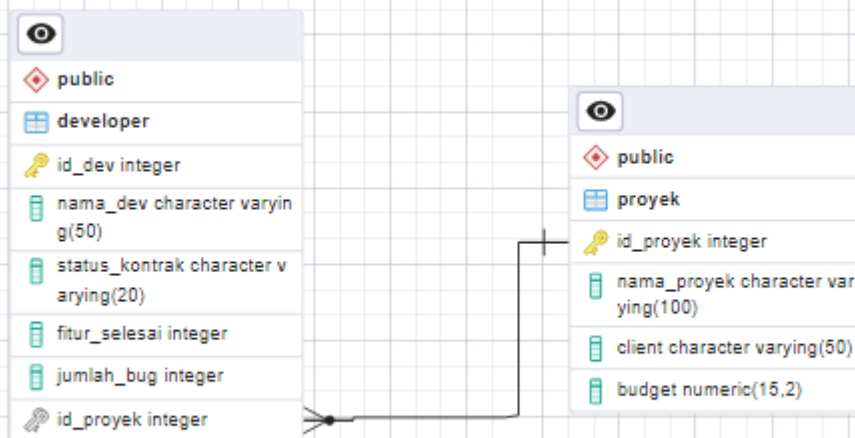
    public double SkorTotal
    {
        get { return _skorTotal; }
        private set { _skorTotal = value; } // Read-only dari luar class
    }
}
```

## Inheritance

```
// Child - Developer mewarisi BaseModel
public class Developer : BaseModel
{
    ...
}

// Child - Proyek mewarisi BaseModel
public class Proyek : BaseModel
{
    ...
}
```

Screenshot ERD Databases pada PgAdmin (klik kanan databases -> ERD For Database):



### Implementasi Kode CRUD:

```
92 private void InsertButton_Click(object? sender, EventArgs e)
93 {
94     if (!ValidateForm()) return;
95
96     var developer = CreateDeveloperFromForm();
97
98     if (_repository.Insert(developer))
99     {
100         MessageBox.Show("Data berhasil ditambahkan!", "Sukses", MessageBoxButtons.OK, MessageBoxIcon.Information);
101         ClearForm();
102         LoadDataGrid();
103     }
104 }
```

```

106     private void UpdateButton_Click(object? sender, EventArgs e)
107     {
108         if (_selectedDeveloperId <= 0)
109         {
110             MessageBox.Show("Pilih data yang ingin diupdate dari tabel!", "Peringatan", MessageBoxButtons.OK, MessageBoxIcon.Warning);
111             return;
112         }
113
114         if (!ValidateForm()) return;
115
116         var developer = CreatedDeveloperFromForm();
117         developer.Id = _selectedDeveloperId;
118
119         if (_repository.Update(developer))
120         {
121             MessageBox.Show("Data berhasil diupdate!", "Sukses", MessageBoxButtons.OK, MessageBoxIcon.Information);
122             ClearForm();
123             LoadDataGrid();
124         }
125     }
126
127     private void DeleteButton_Click(object? sender, EventArgs e)
128     {
129         if (_selectedDeveloperId <= 0)
130         {
131             MessageBox.Show("Pilih data yang ingin dihapus dari tabel!", "Peringatan", MessageBoxButtons.OK, MessageBoxIcon.Warning);
132             return;
133         }
134
135         var result = MessageBox.Show("Apakah Anda yakin ingin menghapus data ini?", "Konfirmasi",
136             MessageBoxButtons.YesNo, MessageBoxIcon.Question);
137
138         if (result == DialogResult.Yes)
139         {
140             if (_repository.Delete(_selectedDeveloperId))
141             {
142                 MessageBox.Show("Data berhasil dihapus!", "Sukses", MessageBoxButtons.OK, MessageBoxIcon.Information);
143                 ClearForm();
144                 LoadDataGrid();
145             }
146         }
147     }

```

## TUGAS OPSIONAL

Screenshot Konsep OOP (Polimorfisme dan Abstraksi):

Polymoprhism

```

18     // Child class untuk Full Time (Polymorphism)
19     2 references
20     public class SkorFullTime : Skor
21     {
22         1 reference
23         public SkorFullTime(int fitur, int bug) : base(fitur, bug) { }
24
25         1 reference
26         public override double HitungSkor()
27         {
28             // Rumus: 10 x Fitur - 5 x Bug
29             return (10 * Fitur) - (5 * Bug);
30         }
31     }

```

Abstraction

```

1 namespace Responsi2_Junpro.Models
2 {
3     // Abstrac class untuk hitung skor (Abstraction)
4     public abstract class Skor
5     {
6         protected int Fitur;
7         protected int Bug;
8
9         public Skor(int fitur, int bug)
10        {
11            Fitur = fitur;
12            Bug = bug;
13        }
14
15        public abstract double HitungSkor();
16    }

```

Screenshot Fungsi pada Postgres:

Functions (12)

- (=) cek\_budget\_tersedia(p\_nama\_proyek character varying)
- (=) count\_dev\_in\_proyek(target\_id\_proyek integer)
- (=) delete\_developer(p\_id\_dev integer)
- (=) get\_all\_developers()
- (=) get\_all\_proyek()
- (=) get\_budget\_proyek(p\_nama\_proyek character varying)
- (=) get\_developer\_by\_id(p\_id\_dev integer)
- (=) get\_id\_proyek\_by\_nama(p\_nama\_proyek character varying)
- (=) get\_proyek\_by\_id(target\_id integer)
- (=) get\_total\_pengeluaran\_proyek(p\_nama\_proyek character varying)